



UNIVERSITI TUNKU ABDUL RAHMAN

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Lee Kong Chian Faculty of Engineering & Science (LKC FES)

UECS1104 Programming and Problem Solving

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Marking scheme

PART A MARKING SCHEME	MARK	MARK OBTAINED
1. Input data design & description	10	
2. Output data design & description	10	
3. Program file & description of details	10	
4. Data memory, formulae & description	10	
5. Structure chart & flow chart	10	
TOTAL	50	

PART B MARKING SCHEME	MARK	MARK OBTAINED
1. Comments/documentation	10	
2. Execution/operations	10	
3. Functionality	10	
4. User friendliness	10	
5. Error free	10	
TOTAL		

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INPUT & OUTPUT DATA DESIGN AND DESCRIPTION

- FOOD ORDERING SYSTEM USER MANUAL

1. Introduction

Welcome to our food ordering system, This user manual will guide you through the process of using our platform to order delicious food from your favorite restaurants.

2. Browsing menu

You will be presented with a table of available food on your order page.

3. Placing order

Once you have found the food you want to order, just input the code of the food, and you can see the food is in your trolley.

If you want to add on other foods, just input 'Y' or 'y' to continue ordering. If not, the system will turn to the payment page.

4. Payment

Review your order total. You will be required to make payment at the counter. After making payment, you will receive a confirmation that your order has been successfully placed.

5. Order tracking

After placing your order, you can see how much time it takes to prepare. You will receive updates on the preparation and estimated completion time of your order.

The user will see the menu display on the output. Our food order system consists of 3 categories, which are food, beverage, and dessert. Each food has attributes such as code, quantity, price and food name.

Food: Nasi Lemak (F001), Roti Canai (F002), Fried Rice (F003).

Beverage: Apple juice (B001), Orange juice (B002), Mango juice(B003).

Dessert: Cheesecake (D001), Ice cream (D002) and Waffle (D003).

The user should be able to input the dish code (e.g., F001) to order and insert the quantity.

Each food code serves as a unique identity. We have configured the codes such that customers may still place orders even if they do not use capital letters. The system will display an error message to the user reminding them to enter the correct dish code if they enter the wrong one. An error message will show up and the user will be prompted to exit the app and place a new order if they enter a negative quantity.

The user may view what food is already in their cart by entering the dish code and quantity. The system will then ask whether they would want to make any changes or add on to their order by input 'Y' , 'y' or 'N', 'n' ,before displaying the final result.

The system will compute the quantity of food with the fixed prices already defined on top and compute the estimated time of preparation. This step won't display to the user.

The system will reset that particular order to 0 if the consumer enters the same dish code again, at which point they can enter the quantity they want again. If there is no need to change their order, the estimated preparation time will be displayed after the order is finished. When the order is finished, the payment and number of orders will be recorded in daily sales overall and daily orders overall, and they will be recorded in a 'daily_statistics.txt' text file. They will accumulate and be updated until the next order is done.

PROGRAM FILE AND DESCRIPTION OF DETAILS

1. Menu Data Text File Format:

- This text file would likely store the details of the menu items available for ordering.
- Each line in the text file could represent one menu item, containing fields separated by a delimiter like a comma or tab.
- Each line represents a menu item with a unique code(String), name(Food), and price(Double, for float).
- Example format:
F001, Nasi Lemak, 9.50
F002, Roti Canai, 3.00
F003, Fried Rice, 6.50
B001, Apple Juice, 8.00
B002, Orange Juice, 7.50
B003, Mango Juice, 9.50
D001, Cheesecake, 7.50
D002, Ice Cream, 4.50

2. Order Data Text File Format:

- This text file could be used to store order details after the customer has selected items.
- Each line in the text file would likely represent one item ordered, with its quantity and calculated price.
- Each line contains the item code, name, quantity ordered, and the total price for that item.
- Example format:
F001, Nasi Lemak, 2, 19.00
B002, Orange Juice, 1, 7.50
D002, Ice Cream, 3, 13.50

3. File Usage and Description:

File Opening and Closing

- The system reads the record of daily sales overall and daily orders overall from the 'daily_statistics.txt' text file using file input/output operations in C++ ('fstream'). Including the file stream objects used and the file paths. When the program starts, the system reads the customer's last accumulated payment history.

File input and output files

- In C++, the system uses 'iostream' to let the user use 'cin' with stream operators (>>), to input the dish code, quantity, and character to make a selection. Output with using 'cout' with stream operators (<<).

4. Implementation Considerations:

- Ensure proper error handling for file operations.
- Use structured data parsing to read and write to the text files.
- Validate and sanitize input/output to avoid malformed data or security vulnerabilities.

DATA MEMORY, FORMULAE AND DESCRIPTION

Data Storage

Global Variables:

- daily_sales_overall: A global variable used to record the total daily sales and display on daily statistics.
- daily_orders_overall: A global variable used to store the number of orders and display on daily statistics.
- total_orders: A global variable to keep track of the total number of orders.

Other local variables used within functions, such as nasilemak_quantity, roticanai_quantity, etc., to store user-inputted order quantities.

Formulas and Algorithms

Formula for calculating total price:

$$\begin{aligned} \text{total_price} = & (\text{fabs})(*\text{nasilemak_price}) + (\text{fabs})(*\text{roticanai_price}) + \\ & (\text{fabs})(*\text{friedrice_price}) + (\text{fabs})(*\text{applejuice_price}) + (\text{fabs})(*\text{orangejuice_price}) + \\ & (\text{fabs})(*\text{mangojuice_price}) + (\text{fabs})(*\text{cheesecake_price}) + (\text{fabs})(*\text{icecream_price}) + \\ & (\text{fabs})(*\text{waffle_price}); \end{aligned}$$

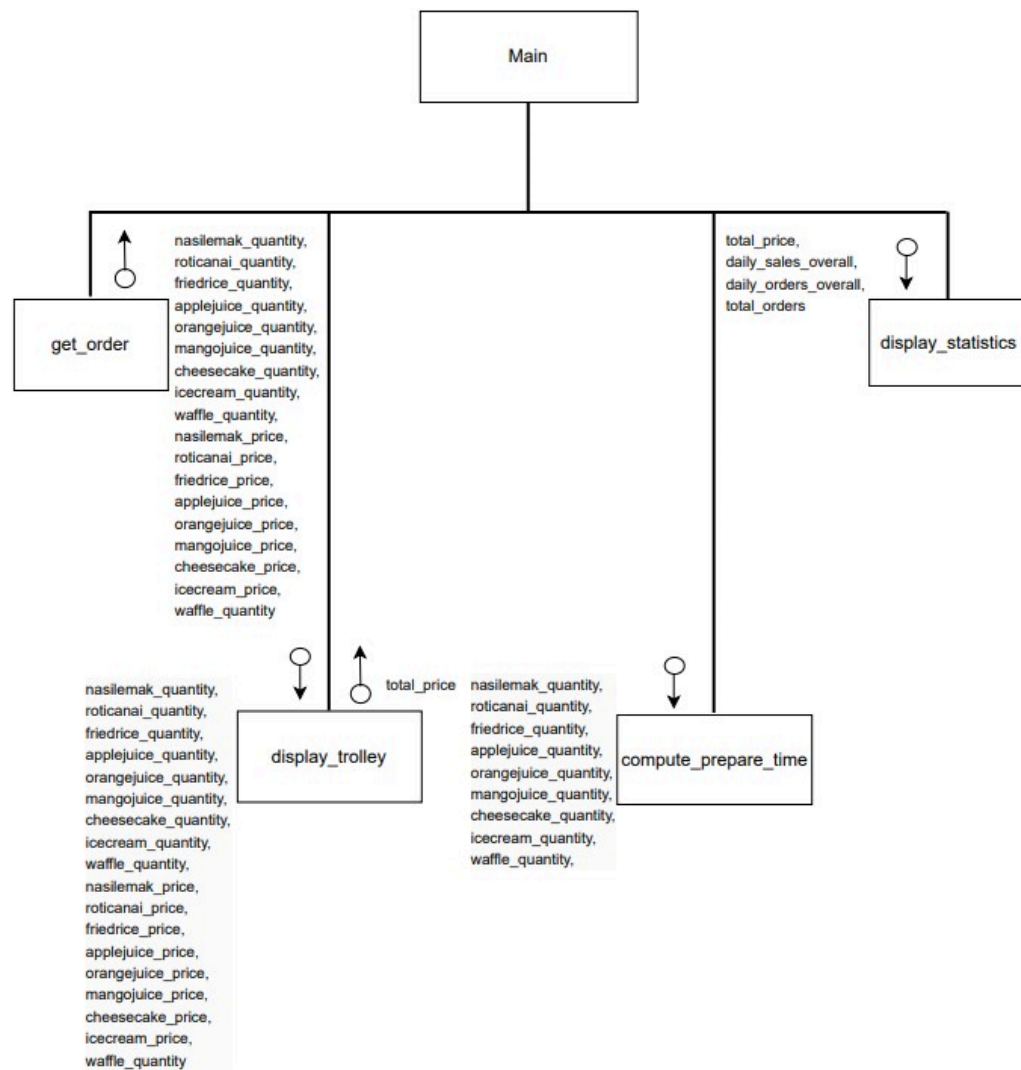
Function ‘fabs’ is used to calculate the absolute value of a floating-point number. It is located in the <cmath> header. Adds the costs of all food items, beverages, and desserts in the basket to get the total shopping cost.

Algorithm for computing preparation time:

```
food_prepare_time = (*nasilemak_quantity + *roticanai_quantity + *friedrice_quantity) * 5;
beverage_prepare_time = (*applejuice_quantity + *orangejuice_quantity +
*mangojuice_quantity) * 2; // Calculate preparation time for beverages
dessert_prepare_time = (*cheesecake_quantity + *icecream_quantity + *waffle_quantity) *
2;
```


Calculates the preparation time depending on the number of various food items, drinks, and desserts requested, in minutes.

STRUCTURE CHART



FLOW CHART

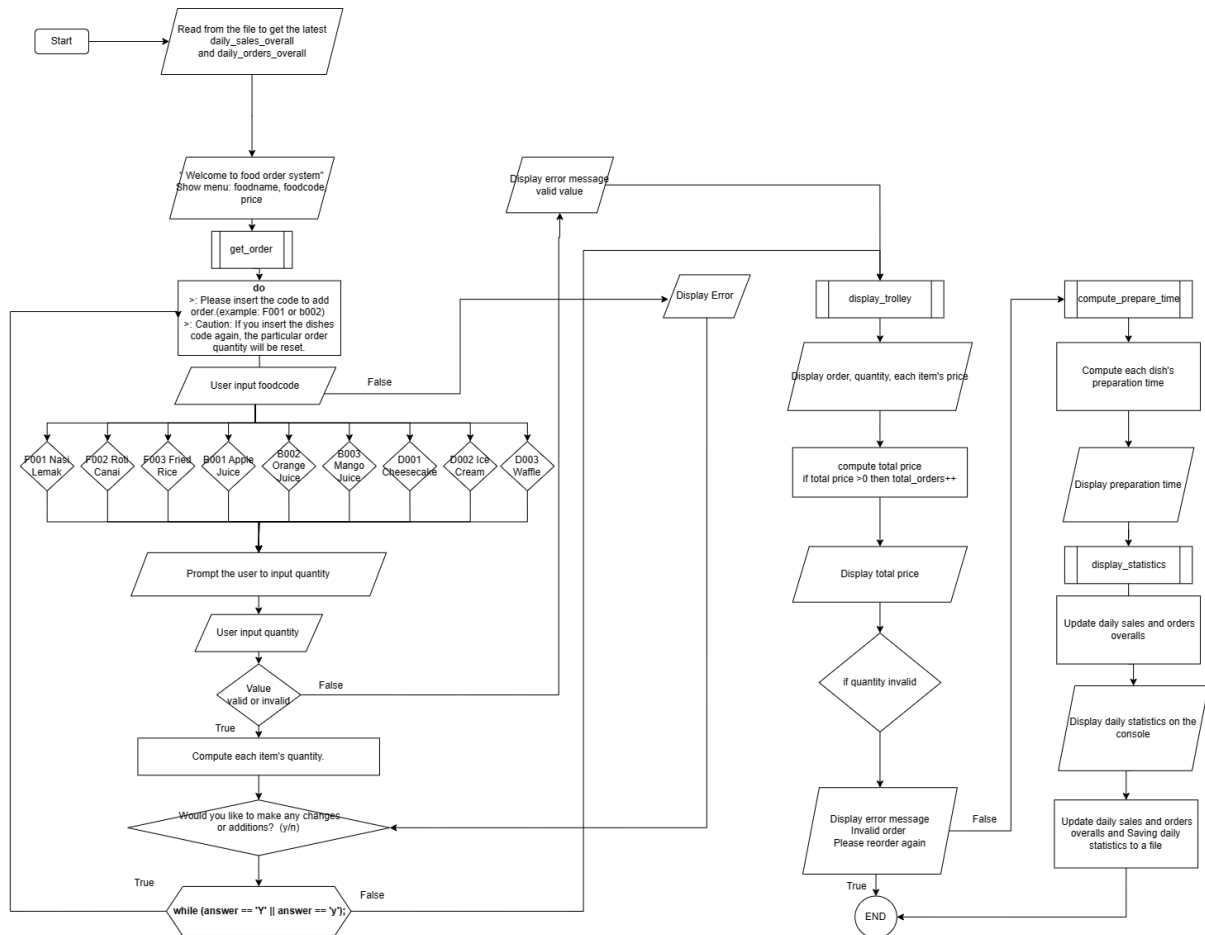


TABLE OF CONTRIBUTIONS

<u>NAME</u>	<u>STUDENT ID</u>	<u>PART</u>
1. CHIN WE JIAN	2204263	-Input & Output data design and description. -Program file & description of details
2. KUAN QING XIANG	2206316	-Input & Output data design and description. -Program file & description of details
3. TEE ZHI YING	2300955	-Data memory, formulae & description -Structure & flow chart -C++ program of food ordering system
4. TING WAI LAM	2301099	-Data memory, formulae & description -Structure & flow chart -C++ program of food ordering system